

COLOSSEUM · SOLANA

# Vitals

A patient is dying on a clock and you decide what happens next.

Anyone can play. Nobody can fake the replay.

EP1 · THE LAST BITE

## THE PROBLEM

## Skill has no passport.

Since 2024, ECFMG certification — the gate to US residency — requires graduating from a school accredited by a WFME-recognised agency. Accreditation is granted to *institutions*, country by country, and most countries do not have it.

**4,350+**

medical schools in the World Directory

**~23**

countries with a WFME-recognised agency

**130–147**

countries whose graduates apply

Every one of those graduates ran hundreds of scored clinical encounters that would have proved what they can do. All of it sits in one institution's database, unverifiable by anyone outside it, and is thrown away at the border. **The evidence exists. The trust layer does not.**

GAMEPLAY & VERIFIABLE OUTPUT

# The replay is the trophy, not the score.



GHOST · MOCKUP

you · adrenaline 1:12 ghost · 0:38

**Ghost racing.** The tape replays exactly, so another player's run can run beside yours — and you watch the two patients diverge while you are still deciding. Trackmania's ghost car, in an emergency room.



RECORD · MOCKUP

harm · stand/walk collapse 50 saves · 12 harms

**Harm is on the record.** Fifty saves with twelve harms reads differently from thirty with none — and it keeps the fastest run and the cleanest run from being the same run.



DEATH CARD · MOCKUP

DeathArrest · turned at 2:14

gold path → adrenaline IM at 0:40

**Failure is content.** "She died because you reached for antihistamines" is a better clip than any win. A loss should be legible, teach in the same breath, and be worth showing someone.

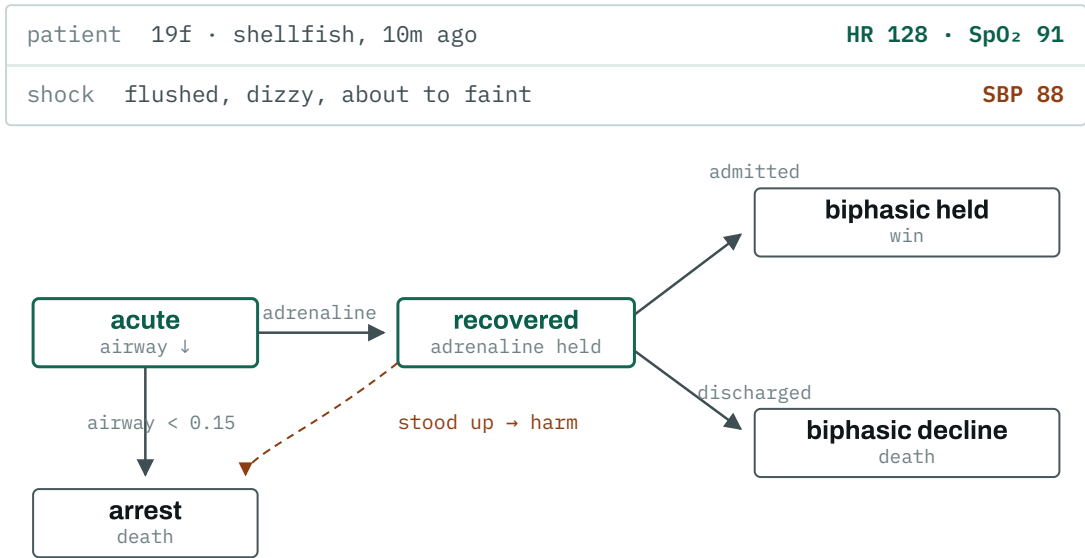
|                                                                                 |                                       |
|---------------------------------------------------------------------------------|---------------------------------------|
| treated adrenaline IM · O <sub>2</sub> · supine · fluids · admit                | WinDischarge · leaf: 65c56e...        |
| stood up adrenaline IM · stood the patient up · O <sub>2</sub> · fluids · admit | WinDischarge · harm · leaf: a6bec3... |
| no adrenaline antihistamine · steroid · wait                                    | DeathArrest · fatal                   |

**Deterministic execution:** Three players, three tapes, replayed against the shipped physiology automaton. Same outcome as the first, different leaf — because harm is on the record.

THE MECHANISM & AUTOMATON

# The model makes it playable. The automaton makes it provable.

THE SITUATION & AUTOMATON



DETERMINISTIC REDUCTION

- 1 · **commit** hash(scenario || player || nonce) onchain
- 2 · **play** Automaton tracks vitals, triggers, harm
- 3 · **reduce** Discrete facts: ordered beats, harm, outcome
- 4 · **anchor** One compressed leaf: recomputed forever
- leaf = sha256( sce\_hash || tape || **beats** || **harm** || **outcome** )

THE AGENT

Plays patient dialogue. In experience, never in evidence.

THE AUTOMATON

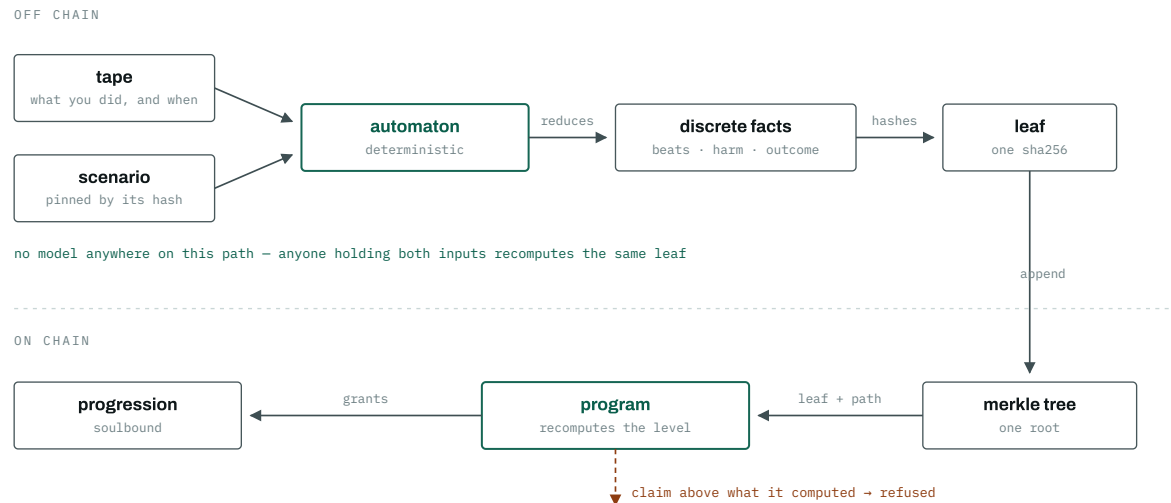
Decides transitions and endings. Exact and replayable.

THE PROOF

No model in proof path. Settles by arithmetic via \$VIGIL.

## ZERO-TRUST ARCHITECTURE

# Two inputs in, one hash out. The program refuses its own user.



The tape and the scenario are the only inputs. The chain never sees a run — only a leaf that has to prove against a root it built itself.

LIVE VALIDATOR · CLAIM: PROFICIENT

REJECTED — claimed Proficient, computed Competent (3 distinct attempts)

LIVE VALIDATOR · CLAIM: COMPETENT

GRANTED — stored Competent, xp 35. Stored level is what program computed.

No issuer key. No oracle. The Solana program recomputes the level from Merkle proofs and writes only what its own arithmetic agrees with.

ONCHAIN ARCHITECTURE

# Four primitives, none of them invented here.

| LAYER             | WHAT IT DOES                                                                        | SOLANA PRIMITIVE                  |
|-------------------|-------------------------------------------------------------------------------------|-----------------------------------|
| Scenario registry | Authors publish; content stays off-chain, only hash + price + royalty split onchain | Anchor program · Token-2022       |
| Replay anchor     | One compressed leaf per run — the high-volume write                                 | Bubblegum state compression       |
| Progression       | Levels, skill trees, badges — minted permissionlessly, predicate recomputed onchain | Token-2022 <i>NonTransferable</i> |
| Credential        | Accredited issuer attests an EPA competency to the player's wallet                  | Attestation Service               |

VOLUME

A single cohort year across the World Directory is hundreds of millions of runs. Compression puts that at **~\$110 per million**. Anchoring only the exams would be the cheap-chain compromise and would kill the thesis.

LATENCY

The skill tree advances in the same screen as the outcome. On a slow chain that becomes a pending spinner and stops feeling like a result.

ONBOARDING

Fee-payer relay. A player never sees a seed phrase, never holds SOL, never learns any of this is happening.

# The token secures verification. It never represents competence.

Someone has to replay the tape and attest the outcome. Make that one party and you have rebuilt the database we are trying to escape. **\$VIGIL is what makes verification permissionless and expensive to lie in.** A vigil is kept beside the patient, never of the patient — which is exactly where this token sits relative to the credential.

## THE LOOP

### STAKE

Operators bond \$VIGIL to accept replay work. Bond size sets how much throughput they can take.

### VERIFY

Replay is deterministic, so **two honest verifiers must agree**. Fees are paid by institutions and sponsors — in USDC, never by players.

### DISPUTE → SLASH

Disagreement is provable: anyone posts a bond and re-runs the tape. The wrong verifier is slashed, the challenger is paid from it. Fraud proofs work here because the computation is exact.

## SINKS · SOURCES

|        |                                                  |            |
|--------|--------------------------------------------------|------------|
| sink   | verifier bond (locked while active)              | supply out |
| sink   | scenario listing + dispute bonds                 | supply out |
| sink   | slashing — burn share                            | burned     |
| source | verification fees → operators                    | earned     |
| source | replay royalties → scenario authors              | earned     |
| note   | authoring is cheap, so the author market is real | —          |

### NEVER TOKENIZED

- **Credentials, badges, skill trees.** Token-2022 NonTransferable, no exceptions — a tradeable "Expert in Cardiology" is credential fraud with extra steps.
- **Access to play.** No token gates a student out of practice, ever. Players never hold \$VIGIL and never see it — the arena has its own name, **3R**, so the two layers do not even share a word.

Allocation is a proposal, not a commitment, and no token ships inside the hackathon window: the verification market only has to exist once verification is worth attacking. Distribution weighted to operators and scenario authors — the two parties whose work the network actually consumes.

# This shape already won this hackathon once.

FRONTIER 2026 · GRAND CHAMPION

**CrowdBrain** — "a vertically-integrated robotics DePIN built to **train users in simulation, qualify them through QA, and route the best operators to real robots.**"

Train in simulation → verify competence → route the qualified to real work. That is Vitals, for robot operators instead of clinicians. We think the pattern generalises; the largest crypto hackathon ever run agreed three months ago, out of 2,857 submissions from 150 countries.

HOW EPISODE SIX GETS MADE — SKALD

A producer console. Type a premise and a number of episodes and an AI showrunner drafts the series bible and scaffolds every storyboard. Then each shot runs a loop: write the prompt, generate, **a crew fails the shot before the video model ever sees it** — a medical advisor on the procedure, a continuity supervisor on the face — approve, and every generation is kept as a take you can roll back to, like a film set. A budget gate blocks spending past its ceiling.

Same principle as the rest of this: something that must be right, checked automatically. **Skald verifies the content. The chain verifies the play.**

WHY MEDICINE FIRST

The hardest instance: highest stakes, heaviest regulation, most protective of data. Solve it here and it generalises downward to every other skill credential.

WHO PAYS

Institutions on ordinary invoices, sponsors funding outcome-linked scholarships, and platform fees on the scenario market. Players pay nothing and hold nothing.

HEAD START

A clinical simulator already in production — 424 authored scenarios, deterministic scoring, a hash-chained audit log, real students.

TEAM

Ex-Medical Sim Lead + Solana Core  
Rust Builder. 424 scenarios shipped.



## WHERE IT STANDS, AND WHAT'S NEXT

# Two halves built. One week to join them.

## WORKING NOW

- **Replay** → **leaf**. EP1 runs against the shipped automaton; three tapes, three distinct leaves, reproducible across runs.
- **Program** → **refusal**. Native Solana program live on a validator, recomputing levels and rejecting claims that don't survive its own arithmetic.
- **Rust end to end** — one scoring implementation compiled into the game, the verifier, and the program.

## THE REMAINING FOUR WEEKS

- **W1** — join the halves: `claim_progress` takes merkle proofs against anchored leaves instead of attempts handed to it.
- **W2** — compressed anchors, scenario registry, public verify page.
- **W3** — soulbound progression, badge-gated scenarios, sponsor escrow paying out on a provable level.
- **W4** — mainnet-beta, demo video, open-source release.

## Cut on purpose

ZK selective disclosure · verifier DePIN with live staking · token launch · mobile · new clinical content. Each is a real v2 line and none survives four weeks. The credential layer stays a stub until an accrediting institution is standing next to it — progression doesn't need one, which is why progression is the demo.

**The 60-second version:** you play a patient who is dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.

EP1 · SHE GOES HOME